

AN2DL - First Challenge Report

soft98ir

Ali Mehdipour, Mobina Mobaraki, Seyed Taha Mousavi Jafarkolaei, Mohammad Zavvari

ALI MEHDIPOUR, Mobina Mobaraki1209, Taha Mousavi, AurelioMind

273930, 274556, 277300, 277925

November 17, 2025

1 Introduction

Estimating pain intensity from wearable sensors is challenging due to noisy signals and strong variability across individuals. This project aims to classify pain levels into three categories using *multivariate time-series data* together with *static contextual features*.

The goal is to build a model that is accurate, robust, and able to generalise to unseen subjects. Our approach combines **temporal feature extraction**, **sequence modelling**, and **fusion of static and dynamic attributes**, while addressing **class imbalance** and overfitting through **regularisation** and **targeted augmentation** [3].

The final pipeline integrates structured preprocessing, a hybrid neural architecture for sequential data, and Stratified K-Fold cross-validation for reliable evaluation [4]. This setup supports systematic model refinement and ensures consistent performance estimates before producing test predictions. Using **autocorrelation**, some features were dropped to improve robustness [1].

2 Problem Analysis

The dataset contains fixed-length multivariate time-series of joint-angle and physiological measurements, along with static descriptors. The task is to classify each sample into one of three pain levels,

requiring the integration of temporal dynamics with static contextual features. Several challenges shape the modelling strategy:

1. **Class imbalance:** The dataset is dominated by the *no pain* class, making minority pain levels harder to model without balancing techniques [3].
2. **High-dimensional sequences:** According to the challenge description, each sample was expected to contain 180 time steps. However, inspection of the provided CSV files showed that all sequences actually have $T = 160$ time steps with consistent time indices, still yielding relatively long sequences with many channels and increasing overfitting risk.
3. **Heterogeneous features:** Time-series signals and static attributes behave differently and must be processed with separate modules before fusion.
4. **Noisy or incomplete data:** Sensor measurements may contain noise or missing values, necessitating careful cleaning and normalisation.
5. **Generalisation challenges:** Limited data, sequence complexity, and imbalance make robust generalisation difficult.

These points motivate hybrid models that capture temporal patterns, fuse static features, and

handle class imbalance. Augmentation and Stratified K-Fold validation further enhance robustness and reliability [4].

3 Method

The proposed method integrates preprocessing, time-series augmentation, a hybrid neural architecture, and Stratified K-Fold cross-validation to perform pain-level classification from multivariate temporal signals.

3.1 Data Preprocessing

Each sample contains a sequence of $T = 160$ time steps with F joint-related features, together with static attributes describing the subject. Time-series values were median-imputed, static features were represented as boolean flags, and all joint inputs were standardised:

$$\tilde{x} = \frac{x - \mu}{\sigma}.$$

The data were reshaped into

$$X_{\text{ts}} \in \mathbb{R}^{N \times T \times F}, \quad X_{\text{static}} \in \mathbb{R}^{N \times S}.$$

Additionally, *joint_30* was removed due to zero variance (constant value of 0.5), and features affected by detected spikes were filtered out accordingly. For temporal structure analysis, we used the autocorrelation function (ACF) on the joint signals [1].

3.2 Augmentation

To mitigate overfitting, lightweight time-series augmentation was applied during training, including Gaussian noise (jitter), uniform scaling, and small temporal shifts. These transformations preserve sequence semantics while increasing variability.

3.3 Correlation Analysis

We first computed pairwise correlations across all numeric variables to identify redundant features. Columns showing extremely high correlation were considered for removal to reduce multicollinearity. **Autocorrelation** analysis was then applied to joint channels to evaluate temporal relevance and to prune features whose autocorrelation decayed almost immediately, behaving similarly to noise [1].

3.4 Neural Architecture

A hybrid network was designed to model both local motion patterns and long-range temporal dependencies:

- **1D Convolutional Encoder:** Extracts short-term temporal features from the transposed input (B, F, T) using stacked Conv1D, Batch Normalisation and ReLU layers with dropout.
- **Bidirectional GRU:** Processes CNN features to capture forward and backward temporal dynamics, yielding hidden states $h \in \mathbb{R}^{T \times H}$ [2].
- **Temporal pooling:** Combines mean and max pooling over time to obtain a compact sequence representation.
- **Static MLP:** A small fully connected network encodes static attributes into a low-dimensional embedding.

Temporal and static representations are concatenated and passed to a final classifier:

$$\hat{y} = \text{softmax}(Wz + b),$$

where z contains fused temporal and static embeddings.

3.5 Training Objective

To address class imbalance, weighted cross-entropy was used:

$$\mathcal{L} = - \sum_{i=1}^N \alpha_{y_i} \log(\hat{y}_{i, y_i}),$$

where α_{y_i} are class weights derived from training-set frequencies [3].

The model was optimised with AdamW [6], gradient clipping to stabilise training, and a learning-rate scheduler that reduces η on validation performance plateaus. Early stopping based on validation macro F1 further limited overfitting.

3.6 Stratified K-Fold Training and Ensembling

Performance was stabilised via Stratified 5-Fold cross-validation, ensuring proportional class representation in all folds [4]. Each fold trains an independent model; their outputs on the test set are averaged:

$$\hat{y}_{\text{final}} = \frac{1}{K} \sum_{k=1}^K f_k(x).$$

This ensemble improves generalisation and reduces fold-specific variance.

4 Experiments

We evaluated our models on a held-out validation split and via Stratified 5-Fold cross-validation. The primary metric is macro F1, which treats all three pain classes equally and is more informative than accuracy under strong class imbalance [3]. After examining the data distribution and autocorrelation patterns, we concluded that zero values should not be imputed or interpolated, as they represent meaningful observations rather than missing data.

Training used batch normalisation, dropout, **weight decay**, **label smoothing** [7], early stopping on validation macro F1, and oversampling with **weighted random sampling** of minority classes in the training splits [3]. Several architectures were prototyped before converging on the final CNN-BiGRU hybrid with static-feature fusion: Several architectures were prototyped before converging on the final CNN-BiGRU hybrid with static-feature fusion. We started from a baseline 1D CNN with static MLP fusion as an initial reference, then moved to a deeper CNN-BiGRU + static backbone to better capture both local patterns and long-range temporal dependencies. We also tested AE/VAE-based embeddings with a small residual classifier, and compared focal loss (with different γ and class weights) [5] against standard weighted cross-entropy. In practice, focal loss did not consistently outperform weighted cross-entropy under oversampling and ensembling, so cross-entropy was retained as the main training objective.

The ACF quantified each joint signal’s temporal dependency, and its derived metrics—AUC and an autocorrelation-based ranking—were used to identify and filter out joints with weak or poor temporal structure, yielding the final feature set used by our best-performing models [1].

Table 1: Improvements with data preprocessing.

Model	F1
CNNGRU raw	0.83
CNNGRU stratified k-fold + ensemble	0.9396
CNNGRU autocorrelation analysis	0.9554

Table 1 summarises representative configurations and their validation macro F1 scores. Values are indicative of typical performance on the held-out validation split.

These results are consistent with the trends observed in cross-validation, where the ensemble of K-Fold models further stabilises macro F1 and improves robustness with respect to fold-specific variability.

5 Results

On the held-out validation set, our best CNN-BiGRU models with static-feature fusion, oversampling, and Stratified 5-Fold ensembling achieved macro F1 in the range 0.95–0.96 (see Table 1). The confusion matrices indicate near-perfect recognition of *no_pain* and high recall on *low_pain* and *high_pain* despite the strong imbalance. On the Codabench public leaderboard, the final ensemble reaches a macro F1 of about 0.95, confirming good generalisation from validation to test data.

6 Discussion

The final pipeline benefits from three main ingredients: a hybrid temporal-static model (CNN-BiGRU with a static MLP), targeted preprocessing (standardisation, spike filtering, autocorrelation-based feature pruning), and imbalance handling via class weights, weighted random sampling, and ensembling [3]. Together, these choices stabilise training and clearly improve minority-class performance. Remaining limitations mainly concern the small dataset size and residual sensitivity on the *high_pain* class, where a few misclassifications can still impact macro F1.

7 Conclusions

We presented a pipeline for pain-level classification from multivariate time-series and static features. Our main contributions are: (i) a CNN-BiGRU backbone with static-feature fusion, (ii) targeted preprocessing, including autocorrelation-based feature selection and spike removal, and (iii) robust evaluation with Stratified K-Fold cross-validation and ensembling [4]. This combination yields strong and stable macro F1 under severe class imbalance. As future work, we plan to explore attention-based architectures and more advanced imbalance-aware objectives to further improve minority-class performance.

References

- [1] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung. *Time Series Analysis: Forecasting and Control*. Wiley, 5 edition, 2015.
- [2] K. Cho, B. Van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. Learning phrase representations using rnn encoder–decoder for statistical machine translation. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734, 2014.
- [3] H. He and E. A. Garcia. Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9):1263–1284, 2009.
- [4] R. Kohavi. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1137–1145, 1995.
- [5] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.
- [6] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [7] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2818–2826, 2016.